

# Deepak Sathyan

## Publication List

### Articles in Review

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5. **Intensity-Frontier Signals of Warped Extra Dimensions** Aug 2026  
D. Kim, D. Sathyan, A. Verma  
[arxiv.org/abs/2608.20482](https://arxiv.org/abs/2608.20482)
4. **Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering** May 2026  
B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan  
[arxiv.org/abs/2605.08010](https://arxiv.org/abs/2605.08010)
3. **A Baryon and Lepton Number Violation Model Testable at the LHC** Aug 2025  
A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan  
[arxiv.org/abs/2508.21064](https://arxiv.org/abs/2508.21064)
2. **New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis** July 2025  
P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang  
[arxiv.org/abs/2507.01000](https://arxiv.org/abs/2507.01000)
1. **United States Muon Collider Community White Paper for the European Strategy for Particle Physics Update** Mar 2025  
M. Begel et al.  
[arxiv.org/abs/2503.23695](https://arxiv.org/abs/2503.23695)

### Articles in Refereed Journals

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5. **New laboratory constraints on neutrinophilic mediators** July 2024  
P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang  
[arxiv.org/abs/2407.12738](https://arxiv.org/abs/2407.12738) (Phys.Lett.B 868 (2025) 139765)
4. **‘Unification’ of BSM searches and SM measurements: the case of lepton+MET and m<sub>W</sub>** Apr 2024  
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan  
[arxiv.org/abs/2404.17574](https://arxiv.org/abs/2404.17574) (JHEP 02 (2025) 139)
3. **A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET** Oct 2023  
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan  
[arxiv.org/abs/2310.13687](https://arxiv.org/abs/2310.13687) (Phys.Lett.B 855 (2024) 138774)
2. **Energy-peak based method to measure top quark mass via B-hadron decay lengths** Dec 2022  
K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan  
[arxiv.org/abs/2212.03929](https://arxiv.org/abs/2212.03929) (JHEP 06 (2023) 021)

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| <b>1. LHC Signals for KK Graviton from an Extended Warped Extra Dimension</b>                | Aug 2020 |
| K. Agashe, M. Ekhaterachian, D. Kim, D. Sathyan                                              |          |
| <a href="https://arxiv.org/abs/2008.06480">arxiv.org/abs/2008.06480</a> (JHEP 11 (2020) 109) |          |

## Articles in Preparation

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| <b>1. LHC as a Beam Dump: Probing Light Dark Photons</b>                                       | Sept 2026 |
| B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, D. Sathyan                                   |           |
| <a href="https://arxiv.org/a/sathyan_d_1">arxiv.org/a/sathyan_d_1</a>                          |           |
| <b>2. New Probes of Dark Sectors at a Muon Collider</b>                                        | Jan 2027  |
| B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, D. Sathyan                                         |           |
| <a href="https://arxiv.org/a/sathyan_d_1">arxiv.org/a/sathyan_d_1</a>                          |           |
| <b>3. Exploring open systems approach to neutrino oscillations</b>                             | Mar 2027  |
| A. Chandra Shekar, K. Kelly, D. Sathyan, L. Strigari, T. Zhou                                  |           |
| <a href="https://arxiv.org/a/sathyan_d_1">arxiv.org/a/sathyan_d_1</a>                          |           |
| <b>4. LHC Signals for KK Gravitons Producing 8-Particle Final States</b>                       | Mar 2027  |
| K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, D. Sathyan |           |
| <a href="https://arxiv.org/a/sathyan_d_1">arxiv.org/a/sathyan_d_1</a>                          |           |

## Ph.D. Thesis

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| <b>1. Unifying Searches for New Physics with Precision Measurements of the W Boson Mass</b>                                                                  | 2024 |
| D. Sathyan                                                                                                                                                   |      |
| <a href="https://drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9">drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9</a> (Ph.D. Thesis) |      |

## Snowmass2021 Contributions

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| <b>5. TF07 Snowmass Report: Theory of Collider Phenomena</b>                                                                          | Oct 2022  |
| F. Maltoni et al.                                                                                                                     |           |
| <a href="https://arxiv.org/abs/2210.02591">arxiv.org/abs/2210.02591</a>                                                               |           |
| <b>4. Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021</b>                              | Sept 2022 |
| K. Agashe et al.                                                                                                                      |           |
| <a href="https://arxiv.org/abs/2209.11267">arxiv.org/abs/2209.11267</a>                                                               |           |
| <b>3. Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021</b>                       | Sept 2022 |
| T. Bose et al.                                                                                                                        |           |
| <a href="https://arxiv.org/abs/2209.13128">arxiv.org/abs/2209.13128</a>                                                               |           |
| <b>2. Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond</b> | Apr 2022  |
| K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan                                                                              |           |
| <a href="https://arxiv.org/abs/2204.02928">arxiv.org/abs/2204.02928</a>                                                               |           |

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| <b>1. Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models</b> | Mar 2022 |
| K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan                      |          |
| <a href="https://arxiv.org/abs/2203.13305">arxiv.org/abs/2203.13305</a>                                        |          |

## Muon g-2 Articles

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| <b>3. The straw tracking detector for the Fermilab Muon g-2 Experiment</b>                                      | Nov 2021 |
| B.T. King et al.                                                                                                |          |
| <a href="https://arxiv.org/abs/2111.02076">arxiv.org/abs/2111.02076</a> (JINST 17 (2022) P02035)                |          |
| <b>2. Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab</b>  | Apr 2021 |
| Muon g-2 Collaboration                                                                                          |          |
| <a href="https://arxiv.org/abs/2104.03240">arxiv.org/abs/2104.03240</a> (Phys.Rev.Accel.Beams 24 (2021) 044002) |          |
| <b>1. Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm</b>                                | Apr 2021 |
| Muon g-2 Collaboration                                                                                          |          |
| <a href="https://arxiv.org/abs/2104.03281">arxiv.org/abs/2104.03281</a> (Phys.Rev.Lett. 126 (2021) 141801)      |          |